stored on hard disks today. To retrieve a piece of information, you first provide some reference data (say, a keyword), the data is then searched for by its address, track, sector and so on, after which it is compared with the reference data. In holographic storage, entire pages can be retrieved, where the contents of two or more pages can be compared optically (light patterns are compared) without having to retrieve the information contained in them. Imagine what this would imply in the field of data mining and warehousing—massive sets of data values can be compared in fractions of a second!

While this technology points to an almost utopian era for storage, there are a number of roadblocks that need to be overcome before you can buy a 1 terabyte holographic crystal from your local hardware store. The success of holographic storage lies in the ability to accurately focus the reference laser on the exact position within the crystal to retrieve that page of information (*see box, 'Inside a Holographic Data Storage System'*). You would be unable to locate the data if there's an error of even a thousandth of an inch. Also, the crystals used in the fabrication of the storage element need to exhibit very exacting optical characteristics to store the data correctly and there are very few substances that adequately and economically meet these needs. The good news is that scientists indicate that holographic storage systems should be available in the next five years and they would debut with capacities of about 125 GB with a data transfer rate of 40 MBps.

Millipede

Flashback to 110 years ago to the age of the punch cards. This was the first device that could be used to store digital information. It was based on a very simple principle of indentations on cards that were read by a machine and translated into 0s and 1s. Fast forward to today. IBM has recently developed a very similar system but has scaled it down to extremely small dimensions, almost bordering on the realm of atoms! Known as Millipede, this technology is theoretically capable of storing information equivalent to about 25 million books in an area the size of a postage stamp! (*See box, 'The Operational Concept behind Millipede'.*)

One of the most obvious advantages with this technology is the fact that very large storage densities can be achieved in very small areas. Lower power consumption makes it ideal for mobile applications such as handheld computers and cellular phones—your next generation cellular phone would be able to hold a gigabyte of multimedia content and contact information! The main hurdle that lies in the path of commercialisation of this technology is the fabrication of the controllers that go into these chips.

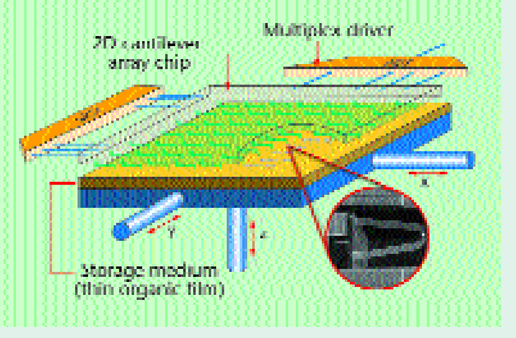
Blue laser

Lasers have long been used in the optical storage of digital data. We see them in everything from CD-ROM to DVD and magneto-optical drives. Information is stored in these devices by encoding data into a stream of 0s and 1s in a spiral on the surface of the media. The amount of digital information that can be stored on a single layer of optical media depends on two factors: the size of the encoded bits on the media surface and how close these bits are packed onto the surface. The density of these bits is directly dictated by the wavelength of the laser beam that reads them. The shorter the wavelength of the light, the narrower is the laser beam, and consequently the smaller can be the size of the bits of information that are encoded on the disk (*see diagram, 'DVD vs Blue Laser CD'*).

As of now, the lasers used in CD and

The Operational Concept behind Millipede

The Millipede chip is created using an array of very tiny cantilevers to create almost atomic-sized indentations in a plastic substrate that is used as the recording material. This array works in a massively parallel fashion where a bank of cantilevers access or create information.



Before a read or write operation, the polymer-based medium, which is just about 50 nanometers thick, is positioned beneath the cantilever array. This medium is mounted on a magnetically driven scanner that can move in three dimensions. During read-write, the medium is moved by the cantilevers along the X-Y axis while the cantilevers actuate and create indentations on the recording surface. Using this process and with a single cantilever design, researchers have managed to achieve a storage density of an astounding 60 to 80 GB per square centimetre. Also, this substrate can be 'erased' and data can be re-written onto it repeatedly. This is achieved by momentarily heating the polymer to a temperature of 150°C so that the surface is effectively smoothened and ready for rewrite. However, individual bits of information cannot be erased; only larger sections of the polymer surface can be cleared. The image above also shows an actual electron microscope image of one of the Millipede cantilevers. The tip of the cantilever head is about 50 Angstroms wide—that's just a few atoms clustered together!